The Null Is Part of the Instrument: Construction-Matched Controls in AI-Assisted Experimental Research

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Abstract

In AI-assisted empirical research, the construction-matched null is not a formality but the primary instrument. We document twelve cases from a three-month research program where the null's design determined the verdict: a positive result survived only when a control specifically capable of killing the claimed mechanism failed to kill it. We propose that null design is a first-class research concern and that the quality of a study is determined by the quality of its nulls, not by the sophistication of its models.

1 Introduction

The standing rule of experimental software research is: a positive result is an artifact until a control specifically capable of killing the claimed mechanism fails to kill it. This rule is easy to state and hard to practice. In AI-assisted research, where agents design, implement, and interpret their own experiments, the temptation to run a token control — one that cannot fail — is strong.

We document twelve cases where the null's design determined the verdict. In six cases, a pre-registered null killed a positive result. In four cases, a control that appeared valid was shown to be invalid by a second test. In two cases, a control that was valid at one model scale became invalid at another.

2 The Null Is the Instrument

Consider the canonical experiment: a system claims to detect misfiled ontology entries at AUC 0.880. The claim is tested against a null. If the null is construction-matched — if it places errors where the system cannot detect them — then the 0.880 is meaningful. If the null is token — if it places errors at random, where any distance-based scorer can detect them — then the 0.880 is a measurement of the null's weakness, not the system's strength.

In our research, the injection policy (random vs. nearest-cell) changed the AUC from 0.880 to 0.506. The system was the same. The metric was the same. Only the null changed. The null is the instrument.

This is not a subtle point. In the perspective-discovery track, the central negative result — cross-embedder ontology agreement is near chance — was established not by a new clustering algorithm but by a new null: conditioning on the stronger support instead of the baseline. The mechanism (D = —s1 - s2— is a folded read-out of s2) was identified by the null, not by the positive result.

3 Twelve Cases

We organize the twelve cases by the phase of research they affect.

3.1 Design Phase

Case 1: Injection distribution (E14). Claim: geometry scores misfilings at AUC 0.880. Null: injection at nearest alternative cell. Result: AUC 0.506. The null killed the claim by changing only where errors were placed.

Case 2: Fixture circularity (E83). Claim: antonym rule detects contradiction. Null: hand-written fixtures without lexical antonyms. Result: recall 0.143 vs. 0.998 on generated fixtures. The null killed the claim by removing the generator's circularity.

Case 3: Adversarial fixtures (E85). Claim: structured rules achieve precision 1.000. Null: adversarial fixtures designed to fire each rule wrongly. Result: precision 0.000. The null killed the claim by inviting the rules to fail.

Case 4: Scale-dependent controls (E19b). Claim: five congruent domains improve abstraction. Null: control arm B5c (five coupled domains, each failing a different clause). Result: the control fires at 4B but is valid at 9B. The null's validity was a property of the stimulus and the model together.

3.2 Implementation Phase

Case 5: Harness parsing (E74). Claim: Cline is a reliable agent. Null: check whether shell actions have output. Result: failure rate 0.005 (parser found no output) vs. 0.070 (parser reads output). The null killed the claim by checking the parser, not the agent.

Case 6: Duplicate sessions (E73). Claim: the corpus has 402 sessions. Null: check for byte-identical session IDs across machines. Result: 188 duplicate records (47% inflation). The null killed the count by checking for duplicates, not by trusting the file count.

Case 7: Truncated embeddings (E78). Claim: revisions form a tight region. Null: check whether the embedding covers the whole file. Result: first 1200 bytes produced identical vectors (within-object distance 0.0000). The null killed the claim by measuring the embedding coverage.

Case 8: Parser measuring the wrong field (E76). Claim: coherence improves after perturbation. Null: check whether the parser reads coherence or node count. Result: parser picked up node count. The null killed the claim by inspecting the parser.

3.3 Interpretation Phase

Case 9: Confound check that stops early (E17). Claim: ANCHORVOC is not circular. Null: three tests (predicts rater's choice; survives half the hints; same on hint-leaking entries). Result: Test 3 fails. The null killed the claim by requiring all three tests to pass.

Case 10: Baseline drift (E87). Claim: pooled baseline scores 0.400. Null: rerun the baseline in the same script. Result: 0.200. The null killed the claim by rerunning, not by trusting the previous number.

Case 11: Aggregation hiding components (E81). Claim: static rules find no structure (ratio 0.96). Null: decompose by cue arity. Result: multi-word cues 15:1, single-word cues 0.92:1. The null killed the claim by decomposing the aggregate.

Case 12: Pre-flight ignored (E10). Claim: the experiment is powered. Null: power analysis. Result: STOP. The run proceeded anyway. The null was overridden by its author — a control the author ignores is not a control.

4 Principles

From these cases we derive three principles:

P1: The null must be construction-matched. A null that places errors where the system cannot detect them is not a control; it is a tautology. The null must perturb the claimed mechanism.

P2: The null's validity must be tested. A control that passes one test and fails another is not a valid control. The workflow must require all pre-registered tests to pass, not just the first reassuring one.

P3: Baselines must be rerun in the same execution. A baseline number cited from a previous run is an assertion about the past, not a measurement. Drift of 0.2 between runs is not rare; it is the norm.

5 Related Work

The importance of separating exploratory from confirmatory analysis is long-established in statistics (Tukey, 1980), and the cost of undisclosed analytic flexibility is well documented in psychology (Simmons et al., 2011). The novel contribution here is the **executable** null: a control that is run in the same script, on the same data, with the same code path, and whose result determines the verdict. In AI-assisted research, where agents can reinterpret results after the fact, the executable null is the only kind that cannot be argued with.

The construction-matched null is related to the adversarial input used to probe machine-learning systems: both are designed to expose the limits of a system rather than to confirm its strengths. The difference is that the construction-matched null is pre-registered: it is specified before the experiment is run, not invented after the result is known. We cite no specific work for the adversarial-input analogy because we have not verified a primary source for it in this revision.

6 Limitations

The twelve cases are from one project, one user, one machine fleet. The frequency of each failure mode in other contexts is unknown. The principles are derived from cases where the null caught the error; we do not have a systematic study of cases where the null failed to catch an error that a human would have caught.

7 Conclusion

The construction-matched null is the primary instrument in AI-assisted empirical research. Its design determines what can be concluded. A positive result is an artifact until a null specifically capable of killing the claimed mechanism fails to kill it. Twelve cases demonstrate that this is not a philosophical preference but a practical necessity.

References

Tukey, J. W. (1980). We need both exploratory and confirmatory. *The American Statistician*, 34(1), 23–25. DOI: 10.1080/00031305.1980.10482706.

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11), 1359–1366. DOI: 10.1177/0956797611417632.